

Epistemic Transfer in Language Models

How SFT on Dialectical Materialism Shifts Causal Reasoning

Mid-Project Peer Review Progress Update

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Motivation

The Problem

- All five frontier AI models recommend carbon pricing for climate policy.
- Evidence is positive but insufficient for the RCP 2.6 warming pathway (below 2°C).
- Stern-Stiglitz target is \$50–100/ton. Global average is \$5.
- Less than 1% of emissions are priced at or above the target.
- Fossil fuel subsidies at \$725B exceed carbon revenue of \$107B by 6.8x.
- None of the five models flag any of these issues.

Our Question

- Can SFT on a non-dominant framework shift reasoning outside the training domain?
- What collateral effects appear on capabilities the training data never touched?

Method

Dialectical Materialism

- A Marxist framework for analyzing social phenomena.
- Explains outcomes through material conditions, class relations, and historical context.
- Produces structural skepticism toward market-based solutions.

What We Did

- Fine-tuned Qwen3.5-9B via QLoRA on 1,500 DM-aligned question-answer pairs.
- Teacher: Qwen3.5-27B generating answers with DM system prompts.
- Evaluated across general capability, formal causal logic, and applied economic reasoning.

Result: the model learns to critique carbon pricing. No other model does this.

11-task evaluation suite: MMLU, HumanEval, GPQA, IFEval, Corr2Cause, EconCausal (4 tasks), and more.

Results: Three Divergent Outcomes

General Capability Preserved

- MMLU: -0.8pp
- HumanEval: 0.0pp
- GPQA: -1.5pp
- No catastrophic forgetting

Formal Causal Logic +38.3pp

- Corr2Cause: 36% to 75%
- 520 errors corrected
- Only 75 new errors
- +81pp on complex templates

Applied Economic Causal -12.4pp

- Task1 Econ: 60% to 48%
- Task1 Finance: 57% to 43%
- Task3: 22% to 11%
- All regressions significant

Root Cause: The Hedging Artifact

The Pattern

- The model converts correct positive causal predictions to ambiguous mixed answers.
- Task1 Economics: 52.7% of regressions are positive to mixed.
- Task1 Finance: 54.6%.

The Source

- The teacher hedges only 4.0% of the time. The pattern is not in the data.
- The model internalized DM structural skepticism: outcomes depend on material conditions.
- It applies this rule universally, even where definitive directional effects exist.

An emergent epistemic prior, transferred through SFT.

The model learned a correct principle (question assumptions) and applied it universally. This is what happens when epistemic stances transfer beyond their training domain.

Implications

What We Learned

- SFT transfers epistemic stances, not just behavioral patterns.
- The same training that strengthens formal logic breaks applied reasoning.
- The model generalized a correct insight into a universal bias.
- Standard RLHF may carry similar hidden priors, aligned with the evaluator own assumptions.

Why It Matters

- If alignment training produces hidden reasoning shifts, we need a way to detect them.
- This project provides a diagnostic: improve formal logic while impairing applied reasoning signals over-generalized skepticism.
- The same diagnostic applies to standard RLHF pipelines.

Future Work (Beyond Final Submission)

The final project submission on June 20 covers Phase 1 only: SFT training and evaluation. The following work continues as ongoing research.

GRPO Training

- Reverse the hedging regression on EconCausal while preserving Corr2Cause gains.
- Two conditions: outcome-only rewards (v3) versus dual advantage with process rewards (v4).
- V3 at step 902 of 1,500. V4 at step 410 of 1,500. V4 leading on outcome reward.

Multi-Ideology Comparison

- Evaluate Liberal and Libertarian SFT models on the same 11-task benchmark.
- Test whether the hedging regression is specific to DM or general to SFT on non-dominant frameworks.
- Models already trained. Evaluation pending after final submission.

Open research question: do standard RLHF pipelines produce similar hidden epistemic transfers, aligned with the evaluator own assumptions?

Timeline: Final Submission June 20

Phase 1: SFT and Evaluation -- Completed

- April to June 2026: Dataset assembly, SFT training, BF16 evaluation across 11 tasks, paper draft.

Phase 2: Final Submission -- June 20

- Final report, slides, and paper covering SFT results, hedging artifact analysis, and implications.

Future Work: GRPO Training -- Ongoing Research

- June to July: Complete v3 and v4 GRPO runs, merge checkpoints, evaluate, compare against SFT baseline.

Future Work: Multi-Ideology Comparison -- Ongoing Research

- July: Evaluate Liberal and Libertarian SFT models. Test whether hedging is DM-specific.

Future Work: Final Analysis -- Ongoing Research

- July to August: Consolidate GRPO and multi-ideology results, statistical testing, extended paper.

Summary

Motivation

- Five models converge on carbon pricing for climate policy, despite evidence of insufficient price levels, net negative fiscal signals, and missing complementary policies.
- Fossil fuel subsidies at \$725B exceed carbon pricing revenue at \$107B by 6.8x.
- Only after SFT on DM-aligned data does the model identify these gaps.

Method

- Fine-tuned Qwen3.5-9B via QLoRA on 1,500 DM question-answer pairs. Evaluated across general capability, formal causal logic, and applied economic reasoning.

Results

- General capability preserved.
- Formal causal reasoning improved by 38 points.
- Applied economic reasoning regressed by 12 points through emergent hedging bias.